

MASTER EXECUTION ROADMAP

Maaden ARGP

Digitalization Platform

Client: Saudi Arabian Mining Company (Maaden)

Project: Ar Rjum Gold Project (ARGP), Kingdom of Saudi Arabia

SoW Reference: ARGP-DIGI-APP-SOW-001 Rev. 00

XDH Reference: XDH-AE-Q26-05-00030

Version 1.0 · 02 July 2026

Executive Summary

The roadmap in five numbers

10 mo

Year 1 Build

W1-W40, 4 phases

7

Lean team roles

PM + Sol Arch + DBA + AI/ML + Sec + Module
+ Doc

93

Tasks in register

Every task with owner + software +
hardware

AED 4.6M

Year 1 cost

Inc VAT + WHT per current XDH quote

AED 8.15M

5-Year TCO

Build + 4 years maintenance

What this roadmap covers

- 5 Modules
CDE, Construction+AWP, Pre-Comm+MC, Quality, Procurement
- 4 Phases per SoW 3.3.10
Requirements (W1-4) | Pilot (W5-12) | Validation (W13-16) | Full Rollout (W17-40)
- 10 enterprise integrations
Aconex, Primavera P6, SAP S/4HANA, AVEVA PI, MS EPM, Azure AD, ADFS, SIEM, Outlook, ServiceNow
- 4 AI Agents per SoW 3.3.4
Progress Verification, RFI Turnaround, Schedule Risk, Punch List Escalation
- 25 KSA compliance items
NCA ECC-1:2018, PDPL, CST CCRF, ISO 27001
- 6 Quality Gates with payment triggers
G1 W4, G2 W12, G3 W16, G4 W24, G5 W29, G6 W40
- Year 2-5 maintenance model
13 support activities, SLA P1-P4, 5-Yr TCO

Client & Project Context

A flagship greenfield mining project for KSA Vision 2030

THE CLIENT

Maaden

Saudi Arabian Mining Company

Established:

1997 (state-owned KSA mining giant)

Business scale:

USD 20B+ annual revenue

Employs 12,000+ staff globally

Listed on Saudi Exchange (Tadawul)

This project sponsor:

Maaden ICT + Maaden PMO

Strategic sponsor: Maaden CIO + CFO

THE PROJECT

Ar Rjum Gold Project

ARGP - flagship greenfield mining + processing

Location:

Ar Rjum, Kingdom of Saudi Arabia

Strategic importance:

Cornerstone of Maaden growth strategy

Key contributor to Saudi Vision 2030

Blueprint for future Maaden mega-projects

Digital scope:

AI-enabled digitalization platform

From design through to startup

Schedule certainty + capital efficiency

Strategic Goals

[Client] Six strategic outcomes per SoW Section 2

01

UNIFIED DATA ENVIRONMENT

Single source of truth for ARGP. Break silos across stakeholders.

02

PREDICTIVE PROJECT CONTROLS

AI/ML for proactive insight into schedule, cost, quality, safety.

03

DIGITAL WORKFACE TRANSFORMATION

Digital work packaging, constraints, real-time progress validation.

04

SEAMLESS DIGITAL HANDOVER

Data-rich transition from construction to commissioning.
Auto MC Dossier.

05

KSA COMPLIANCE + SECURITY

NCA ECC, PDPL, CST CCRF. In-Kingdom hosting. 24/7 SOC monitoring.

06

REPLICABLE BLUEPRINT

Template for future Maaden mega-projects. Modular, scalable design.

5 Core Modules

[Client] The digital platform structure per SoW Section 3

M1

Common Data Environment (CDE) + Engineering Data Hub

Federated Model Viewer, Doc Mgmt, Intelligent P&ID, Master Data Registry

M2

Construction Management + AWP

Digital Work Packaging (EWP/CWP/IWP), Constraint Kanban, Progress Measurement, Timesheets

M3

Pre-Commissioning + Mechanical Completion

Digital Pre-Comm Plan, MC Dossier Assembly, Digital Check Sheets, Punch List Database

M4

Quality Management + Assurance

NCR management, ITP tracking, welding QA, NDT records, supplier quality

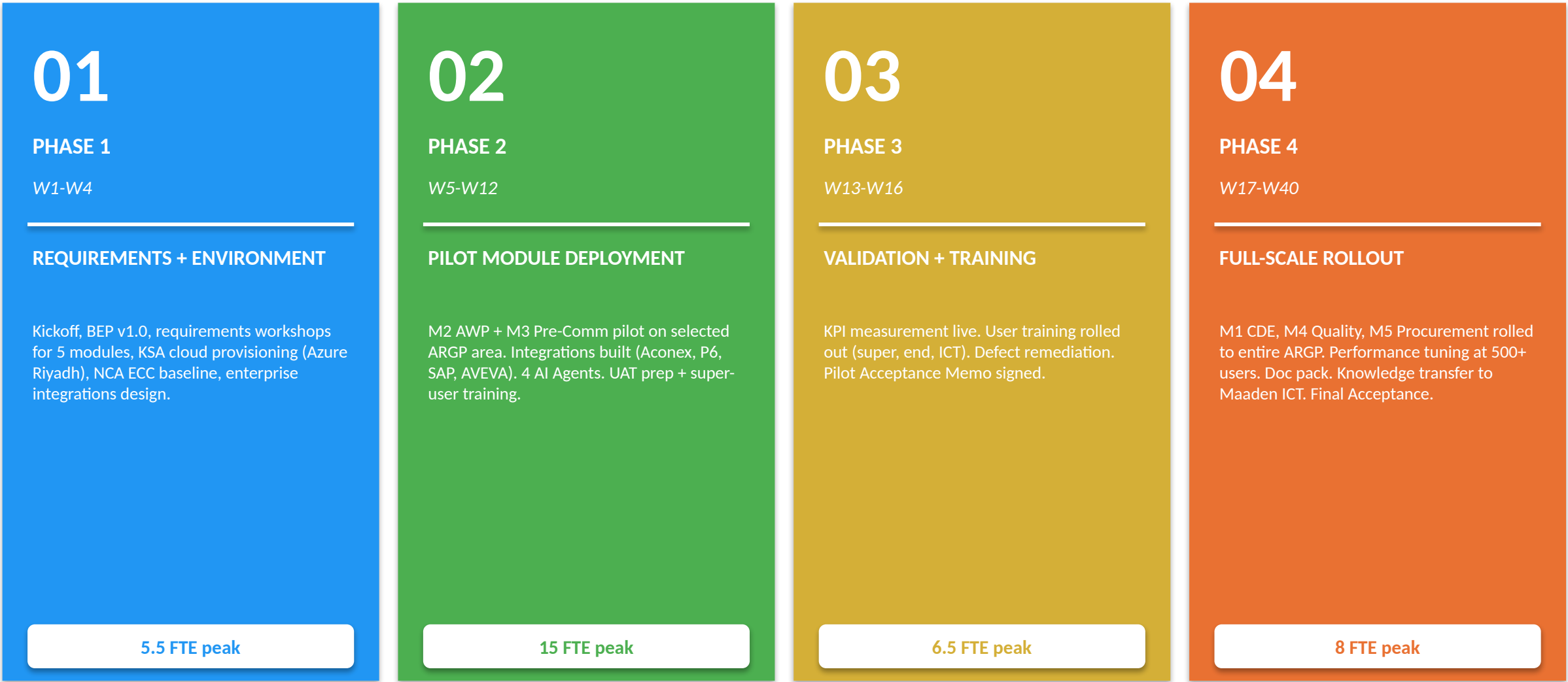
M5

Procurement + Supply Chain Management

MRQ/RFQ workflows, PO tracking (SAP integration), expediting, inspection records

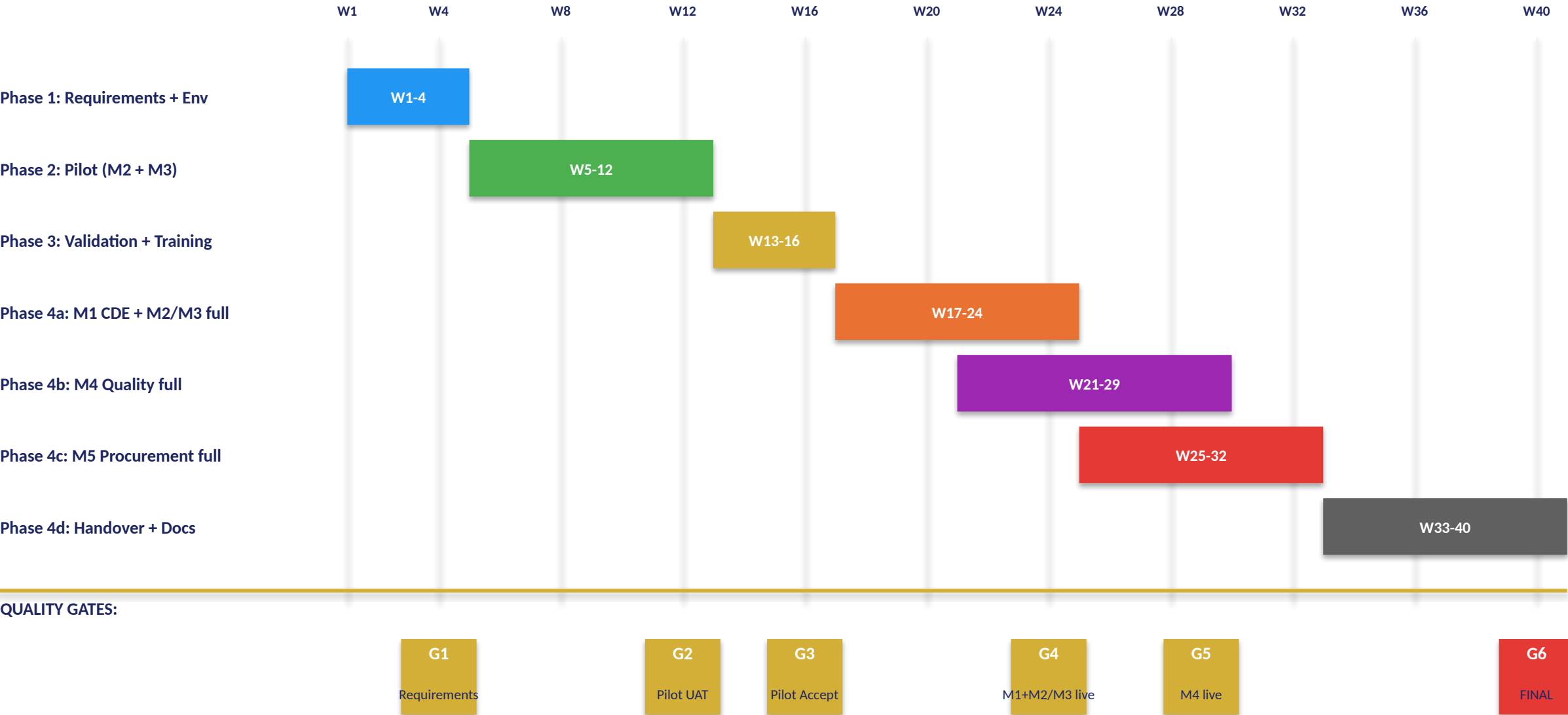
4-Phase Delivery per SoW 3.3.10

[Client] W1-W40 Year 1 build



Year 1 Programme Timeline

[XDH] 40-week Gantt with phase overlaps and quality gates



Team Composition • 7 Lean Roles

[XDH] Total 77 person-months across Year 1



Technology Stack

[XDH proposed] 40+ tools across dev, cloud, AI, integration, security

CLOUD PLATFORM (KSA)

- Microsoft Azure Riyadh
- Azure SQL / Cosmos DB
- Azure Databricks
- Azure Sentinel (SIEM)

AI + ML

- Anthropic Claude API
- Azure OpenAI
- HuggingFace
- MLflow + Databricks

ENTERPRISE INTEGRATIONS

- Oracle Aconex
- Primavera P6 EPPM
- SAP S/4HANA
- AVEVA PI System
- Microsoft EPM
- Azure AD / ADFS SSO

DEV + CI/CD

- GitHub Enterprise + Copilot
- Azure DevOps Pipelines
- VS Code + JetBrains
- Node.js + TypeScript
- React + Next.js + Tailwind

SECURITY + MONITORING

- Azure Sentinel SIEM
- Burp Suite (external pen)
- OWASP ZAP
- Datadog APM
- PagerDuty

BIM / CAD (M1)

- Autodesk Navisworks
- Autodesk Revit
- Autodesk Plant 3D
- APS / Forge Viewer
- IFC.js / xeokit

Enterprise Integrations Map

[Client] 10 systems the platform must integrate with



Protocols: REST/OAuth 2.0, SAP OData, SAML 2.0 for SSO, AVEVA PI Web API. All integrations built in Phase 1-2.

4 AI Agents per SoW 3.3.4

[Client mandated] AI-powered capabilities in the platform

01

Progress Verification

Anthropic Claude + Vision

Ingests IWP completion data + site photos + timesheets. Auto-produces daily progress % per work package.

02

RFI Turnaround Predictor

Claude + XGBoost

Classifies incoming RFIs by priority, predicts likely turnaround time, escalates aged items.

03

Schedule Risk Forecast

Prophet + XGBoost + Claude

ML model on historical schedule + productivity. Forecasts likely completion date, highlights risks.

04

Punch List Escalation

Claude + rules engine

Scores every punch item on age + severity + system criticality. Auto-escalates blocking items.

KSA Compliance Framework

[Client mandated] Four regulatory pillars

NCA ECC-1:2018

Cybersecurity

National Cybersecurity Authority Essential Cybersecurity Controls. IAM, encryption, incident response, SIEM.

Requirements:

10 controls

Sign-off: External auditor annual

PDPL

Data Privacy

Personal Data Protection Law from SDAIA. DPIA required. Data subject rights. Consent management.

Requirements:

8 controls

Sign-off: DPO + attestation

CST CCRF

Cloud Framework

Cloud Computing Regulatory Framework. In-Kingdom hosting mandatory. Level 1-4 classification.

Requirements:

4 controls

Sign-off: CST filing

ISO 27001

Info Security

International standard. Pre-audit in Y1, full certification in Y2.

Requirements:

3 controls

Sign-off: ISO auditor Y2

Total 25 compliance controls tracked in Master Roadmap KSA Compliance sheet.

6 Quality Gates • Payment Milestones

[XDH proposed] Aligned with SoW 3.3.10

G1	W4	REQUIREMENTS SIGN-OFF BEP + Requirements + Solution Arch + Envs live + NCA ECC baseline	15%
G2	W12	PILOT UAT + INTEGRATIONS M2/M3 Pilot live + 4 AI Agents + Enterprise integrations + Pen test	20%
G3	W16	PILOT ACCEPTANCE Pilot validated in Prod + KPIs baselined + Super users trained	15%
G4	W24	M1 + M2/M3 FULL LIVE CDE full live + M2/M3 rolled to all users + Perf SLA at 500+ users	15%
G5	W29	M4 QUALITY FULL LIVE Quality Mgmt full live + Field test + ISO 27001 pre-audit	15%
G6	W40	FINAL ACCEPTANCE M5 Procurement live + Docs + Runbooks + ICT handover + Cert	20%

Payment Distribution • Year 1

[XDH proposed] 8 tranches summing to 100%

PAYMENT BREAKDOWN

Mob	Mobilisation on PO	5%	AED 230K
G1	Requirements Sign-off (W4)	10%	AED 461K
G2	Pilot UAT + Integrations (W12)	15%	AED 691K
G3	Pilot Acceptance (W16)	15%	AED 691K
G4	M1 + M2/M3 Full Live (W24)	15%	AED 691K
G5	M4 Quality Full Live (W29)	15%	AED 691K
G6	Final Acceptance (W40)	15%	AED 691K
Wrap	Final handover (W41)	10%	AED 461K
TOTAL		100%	AED 4.6M

YEAR 1 GRAND TOTAL

AED 4.6M

USD 1.25M

Includes 5% KSA WHT

Hardware + Licences 15%, Margin 40%

PAYMENT SCHEDULE

Invoices raised 20th of each month. Payment: within 30 calendar days.

Risk Heat Map • Top 25 Risks

[XDH proposed] Score = Likelihood x Impact



TOP MITIGATION PLAYS

- 01

KSA CLOUD
Request tenancy Day 1. Escalate to Microsoft Riyadh. Have Oracle + STC fallback.
- 02

ENTERPRISE ACCESS
Raise all ITSM tickets Day 1. Weekly follow-up.
- 03

NCA ECC BASELINE
KSA Security engaged Day 1. External pen test W14 buffer.
- 04

SCOPE CREEP
Firm BEP v1.0 in W4 as baseline. Change control per T&C.
- 05

USER ADOPTION
Super users in W11. User forums. Executive comms.
- 06

AI COST CONTROL
Weekly API spend monitoring. Prompt optimisation.

Deliverables Register • 25 Items

[Client + XDH] Every deliverable per SoW 3.4.5

W4Doc BEP v1.0	W4Doc Requirements v1.0	W4Doc Solution Arch v1.0	W4Infra KSA Cloud envs	W4Doc NCA ECC baseline
W4Doc PDPL DPIA	W12SW M2 AWP Pilot	W12SW M3 Pre-Comm Pilot	W12SW 4 AI Agents	W12SW Enterprise integrations
W15Doc Pen Test v1.0	W16Doc Pilot Acceptance	W13SW KPI Dashboards	W13Trng Super User training	W36Trng End User training
W24SW M1 CDE full	W29SW M4 Quality full	W32SW M5 Procurement full	W38Doc User Manual	W38Doc Admin Manual
W38Doc Tech Docs Pack	W39Doc Operational Runbooks	W39Trng ICT Handover	W32Doc ISO 27001 pre-audit	W40Doc Final Acceptance

16 / 22

Year 2-5 Maintenance Model

[XDH proposed] 4 years of ongoing support post-Y1 handover

SUPPORT TIERS • L1/L2/L3

L1	TIER 1 HELPDESK First-line user support <i>SLA: Response 4h, resolution 24h</i> <i>Availability: Continuous</i>
L2	TIER 2 TECHNICAL Bug diagnosis, config, minor fixes <i>SLA: Response 1h, resolution 1 business day</i> <i>Availability: Bus hrs + escalation</i>
L3	TIER 3 DEVELOPER Code changes for critical bugs <i>SLA: Response 15 min for P1</i> <i>Availability: 24/7 on-call</i>

ANNUAL COST TREND

Y1	Build phase Annual cost AED 4.6M
Y2	Maintenance start Annual cost AED 805K
Y3	Rise 5% Annual cost AED 848K
Y4	Rise 10% Annual cost AED 890K
Y5	Rise 15% + closeout Annual cost AED 932K

5-YEAR TCO

AED 8.15M / USD 2.2M (at AED-USD 3.6725)

Critical Path • Cannot Slip

[XDH proposed] The chain that determines delivery date



IF THESE SLIP, YOU SLIP

Every step above must land on-time

- KSA cloud >W2

Blocks all dev

- Access requests >W3

Blocks integrations

- NCA ECC issue

Blocks pilot go-live

- Pilot UAT fail

Blocks G3 payment

- Integration issue

Blocks Phase 4

- Late scope add

Blocks W40 target

CRITICAL PATH INSIGHT

Every gate week has an accountability owner. Weekly report to Maaden PMO tracks slip risk.

Meeting Cadence • Communication Plan

[XDH proposed] 18 meeting types

DAILY + WEEKLY

- Daily Standup

15 min | All team | Blockers

- Weekly Client Status

Mon 30 min | PM + Maaden PMO

- Security Sync

Wed 30 min | KSA Sec + CISO

- Integration WG

Thu 60 min | Sol Arch + Maaden IT

SPRINT + FORTNIGHTLY

- Sprint Planning

Every 2wk 90 min | PM + Sol Arch + AI/ML

- Sprint Review + Demo

Every 2wk 60 min | All + Maaden PMO

- Sprint Retrospective

Every 2wk 60 min | All XDH

- CCB Change Control

As needed | PM + Maaden PMO

MONTHLY + BEYOND

- Steering Committee

Monthly 90 min | PM + Maaden CIO

- AI Model Review

Monthly 60 min | AI/ML + SMEs

- User Forum

Monthly 60 min from P3

- Executive Briefing

Quarterly 30 min

- Compliance Audit Prep

Bi-annual

Week 1 Kickoff Checklist

[XDH proposed] Day-by-day actions for first 5 working days

DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
☐ Signed PO received	☐ Client kickoff (2h)	☐ Req workshop #1	☐ Req workshop #2	☐ Req workshop #4
☐ XDH team kickoff	☐ Team RACI	☐ Sol Arch v0.3	☐ Req workshop #3	☐ Req workshop #5
☐ Slack channel	☐ ITSM tickets raised	☐ Azure AD SSO req	☐ BEP v0.8 circ	☐ Wk 1 summary
☐ ACC workspace	☐ KSA cloud request	☐ NCA ECC baseline	☐ Sandbox provision	☐ Sprint 1 planning
☐ HW spec drafted	☐ BEP v0.5 draft	☐ Data mapping	☐ Aconex sandbox	☐ Weekend prep

Success Criteria

[XDH + Client] Six metrics for "we won"

01

ON TIME

All 6 quality gates hit within +/- 1 week. Final Acceptance on/before W40.

02

ON BUDGET

Delivered within AED 4.6M lump sum. Variations per T&C.

03

ON QUALITY

Meet all 8 SoW KPIs. Zero P1 defects at Final Acceptance.

04

USER ADOPTION

>80% daily active users of AWP within 3 months of rollout.

05

COMPLIANCE PASSED

NCA ECC passed. PDPL attestation. ISO 27001 pre-audit.

06

CLIENT DELIGHT

Maaden refer XDH to Aluminium, Waad Al Shamal projects.

If we hit 6/6, XDH becomes preferred vendor for Maaden portfolio expansion in Y2-Y5.

Where the detail lives

This deck is the visual summary. The Excel workbook is the working reference.

Maaden ARGP - Master Roadmap.xlsx (14 sheets)

Pipeline\Maaden\2026.05.18 PDE Digitalization\Master Roadmap\

1-2	Cover + Overview · Strategic context, 5 modules, 4 phases, 8 KPIs
3	Programme Timeline · 40-week Gantt with phase, module, integration tracks
4	Task Register (MASTER) · 93 tasks with owner, software, hardware, deliverables
5-7	Team Roster · Software Stack (40+) · Hardware + Infra (20 items)
8-9	Integrations Register (10) · KSA Compliance (25 controls)
10-12	Deliverables (25) · Quality Gates (6) · Risk Register (25)
13-14	Year 2-5 Maintenance model · Meeting Cadence (18 types)